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(a)	The <u>rate of change of angular displacement</u> of the object about the centre of the circle.
(b)(i)	Horizontal component of normal contact force N provides centripetal force. $N \sin \theta = F_c = mr\omega^2$ $N \cos \theta = mg$
	$\frac{\sin \theta}{\cos \theta} = \frac{0.200 \sin \theta \left(\frac{2\pi}{0.75} \right)^2}{9.81}$
	$\theta = 45.7^\circ$ (1dp)
(b)(ii)	<u>Centripetal force will increase</u> thus <u>the horizontal component of the normal contact force will be greater</u> . Bead will move higher and value of θ will increase.
(c)(i)	Resultant force = gravitational force by the Earth + gravitational force by the Sun
	$F_{\text{Earth}} = \frac{GMm}{r^2} = \frac{6.67 \times 10^{-11} (5.98 \times 10^{24}) 6200}{(1.50 \times 10^9)^2} = 1.10 \text{ N}$ $F_{\text{Sun}} = \frac{GMm}{r^2} = \frac{6.67 \times 10^{-11} (1.99 \times 10^{30}) 6200}{(1.50 \times 10^9 + 1.50 \times 10^{11})^2} = 35.9 \text{ N}$
	Resultant force = $1.10 + 35.9$ $= 37.0 \text{ N}$
(c)(ii)	Gravitational force by the Earth and the Sun provides centripetal force. $F_{\text{net}} = mr\omega^2$ $37.0 = 6200(1.515 \times 10^{11})\omega^2$
	$\omega = 1.98 \times 10^{-7} \text{ rad s}^{-1}$ (3sf)
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